Gradient-based dimension reduction

Part 3: Dimension reduction for Bayesian inverse problems

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PILearnWater
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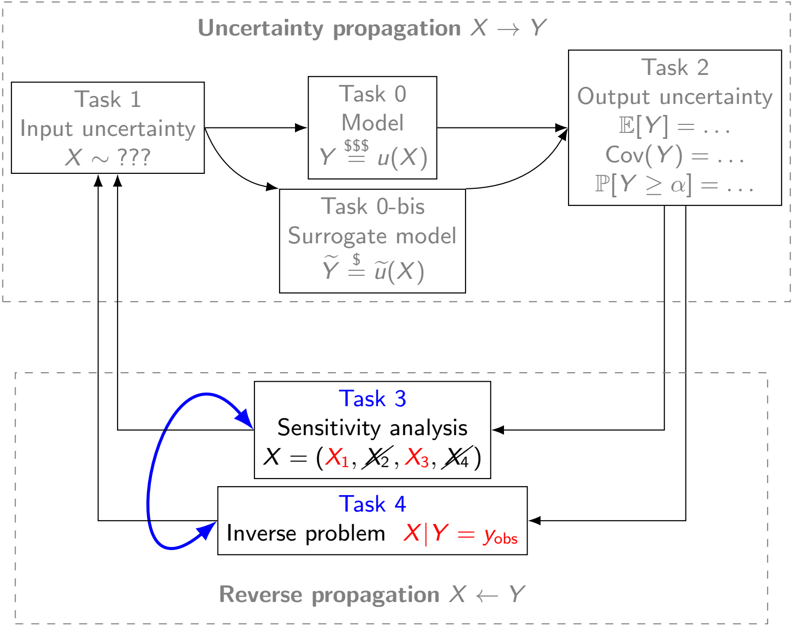


Uncertainty propagation $X \to Y$
Task 1
Input uncertainty
$X \sim ???$
Task 0
Model
$Y \overset{\$\$\$}{=} u(X)$
Task 0-bis
Surrogate model
$\widetilde{Y} \overset{\$}{=} \widetilde{u}(X)$
Task 2
Output uncertainty
$\mathbb{E}[Y] = \dots$
$\mathrm{Cov}(Y) = \dots$
$\mathbb{P}[Y \geq \alpha] = \dots$
Task 3
Sensitivity analysis
$X = (X_1, \cancel{X_2}, X_3, \cancel{X_4})$
Task 4
Inverse problem $X|Y = y_{\text{obs}}$
Reverse propagation $X \leftarrow Y$

Inverse problem

Given an observation y_{obs} of

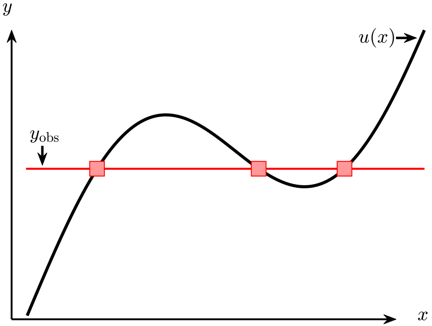
$$Y = u(X_1, \ldots, X_d) + \varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, \Sigma_{\text{obs}})$, how to identify the parameter X which is the most coherent with this observation?

The Bayesian perspective: update the "knowledge" on X after observing y_{obs}.

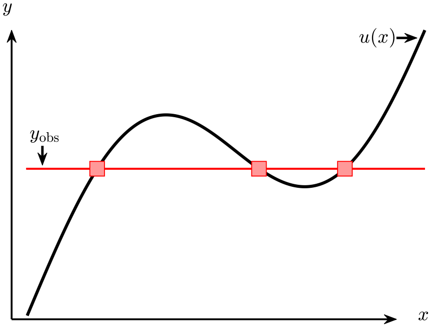
$\pi_X(x) \quad \longrightarrow$  $\longrightarrow \quad \pi_{X|Y}(x|y_{\text{obs}})$

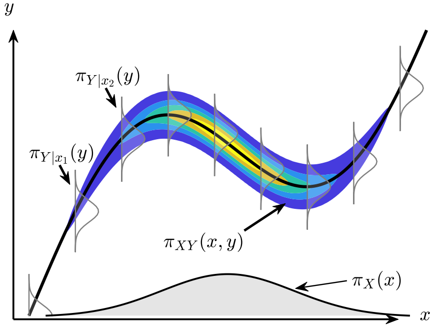
Variational



$$\min_{x} \underbrace{\tfrac{1}{2}\|y_{\mathrm{obs}} - u(x)\|^2_{\Sigma_{\mathrm{obs}}^{-1}}}_{\text{data mismatch}} + \underbrace{\lambda \mathcal{R}(x)}_{\text{regularization}}$$

Variational **V.S.** Bayesian





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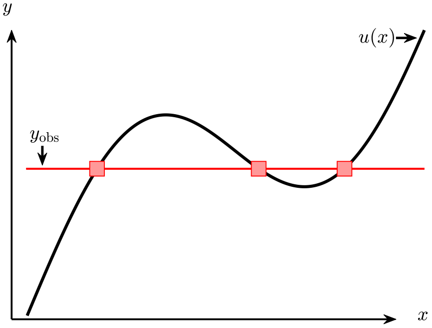

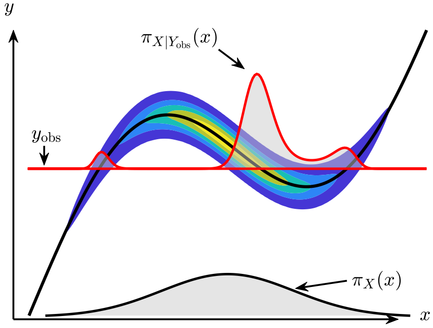

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Given y_{obs}, the **posterior** is given by $\boxed{\pi_{X|y_{\text{obs}}}(x) \propto \pi_{Y|X}(y_{\text{obs}}|x)\pi_X(x)}$

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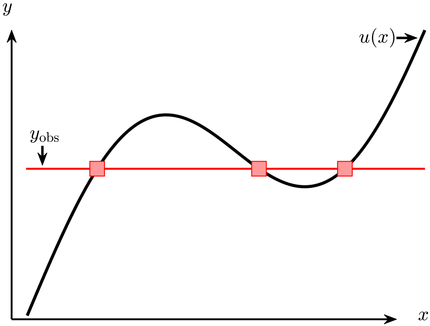

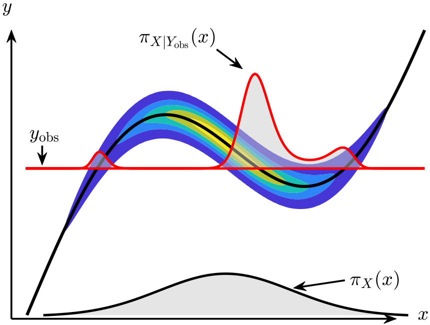

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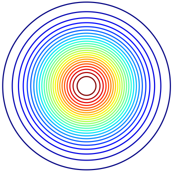
Curse of dimensionality $x = (x_1, \ldots, x_d)$

Standard sampling algorithms (MCMC, Importance Sampling, Particle Filters) **struggle when** $d \gg 1$ (slow mixing, weight degeneracy, slow convergence...)

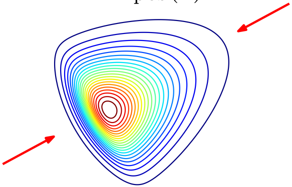
Low-effective dimension of Bayesian inverse problems

In many situations, the data are informative only on a **low-dimensional subspace**

$\pi_{\mathrm{pr}}(x)$



$\pi_{\mathrm{pos}}(x)$



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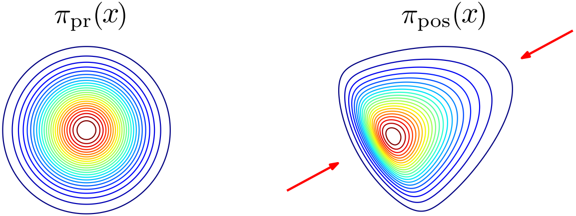

The **likelihood informed** components

$$X = \underbrace{U_m X_m}_{\text{informed}} + \underbrace{U_\perp X_\perp}_{\text{uninformed}}$$

Find a projection matrix $U_m \in \mathbb{R}^{d\times m}$ (**feature map**) and a reduced likelihood $\widetilde{\mathcal{L}}^y : \mathbb{R}^m \to \mathbb{R}_{\geq 0}$ (**profile function**) such that

$$\widetilde{\pi}_{X|Y}(x|y) \propto \widetilde{\mathcal{L}}^y(U_m^\top x)\pi_X(x)$$

yields a "good" posterior approximation to $\pi_{X|Y}(x|y) \propto \mathcal{L}^y(x)\pi_X(x)$

Subspace accelerated inference

[Vrugt 2009] Accelerating MCMC [...] with randomized **subspace sampling**, INT J NONLIN SCI NUM
[Marzouk Najm 2009] Dimensionality reduction and PC acceleration of Bayesian inference, JCP
[Cui Law Marzouk 2016] Dimension-independent **likelihood-informed** MCMC, JCP
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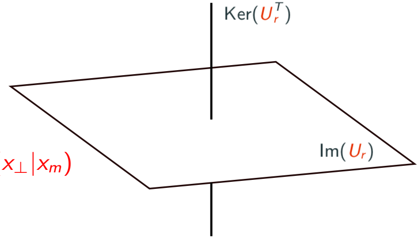

Then

$$\widetilde{\pi}_{X|y_{\mathrm{obs}}}(x) = \underbrace{\left(\widetilde{\mathcal{L}}(x_m)\pi_{X_m}(x_m)\right)}_{\widetilde{\pi}_{X_m|y_{\mathrm{obs}}}(x_m)} \pi_{X_\perp|X_m}(x_\perp|x_m)$$

Concentrate the numerical effort on X_m

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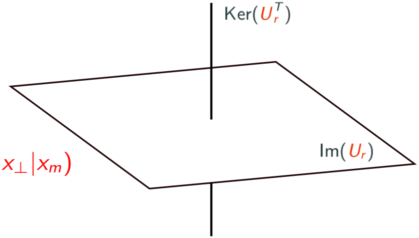

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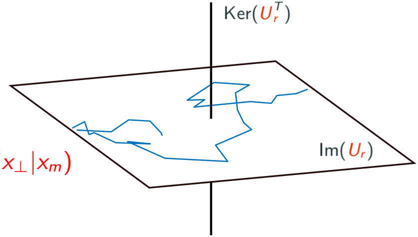

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- Maximum a posteriori $\max_x \pi_{X|Y}(x|y)$ over $x \in \mathrm{Im}(U_m) + U_\perp x_\perp^0$
- MCMC to sample from $\widetilde{\pi}_{X|y}$
  1. **Subspace MCMC** to get samples $x_m^{(i)} \sim \widetilde{\pi}_{X_m|y_{\mathrm{obs}}}(x_m)$

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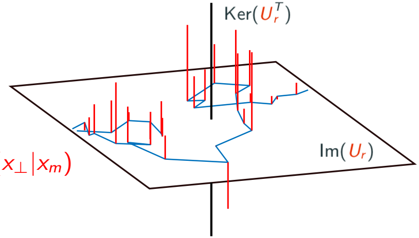

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- MCMC to sample from $\widetilde{\pi}_{X|y}$
  1. **Subspace MCMC** to get samples $x_m^{(i)} \sim \widetilde{\pi}_{X_m|y_{\mathrm{obs}}}(x_m)$
  2. Draw samples from the **conditional prior** $x_\perp^{(i)} \sim \pi_{X_\perp|X_m}(x_\perp|x_m^{(i)})$
  3. Assemble $x^{(i)} = U_m x_m^{(i)} + U_\perp x_\perp^{(i)} \sim \widetilde{\pi}_{X|y}(x)$

Likelihood informed subspace

Gaussian Linear problems: a well understood case

$$X \sim \mathcal{N}(\mu_{\text{pr}}, \Sigma_{\text{pr}})$$
$$Y = \underbrace{u(X)}_{AX} + \varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, \Sigma_{\text{obs}})$. Then

$$(X|Y = y) \sim \mathcal{N}(\mu_{\text{pos}}(y), \Sigma_{\text{pos}})$$

with $\mu_{\text{pos}}(y) =$ (blablabla) and

$$\boxed{\Sigma_{\text{pos}}^{-1} = \Sigma_{\text{pr}}^{-1} + \underbrace{A^\top \Sigma_{\text{obs}}^{-1} A}_{H}}$$

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The prior-to-posterior **covariance contraction** is captured by H, and the most informed direction are given by

$$\min_{v \in \mathbb{R}^d} \frac{v^\top \Sigma_{\text{pos}} v}{v^\top \Sigma_{\text{pr}} v} \quad \overset{(\star)}{\Leftrightarrow} \quad \max_{v \in \mathbb{R}^d} \frac{v^\top H v}{v^\top \Sigma_{\text{pr}}^{-1} v}$$

[Flath et.al. 2011] Fast algorithms for Bayesian UQ in large-scale linear inverse problems [...], SIAM-SISC

[Bui-Thanh et.al. 2012] Extreme-scale UQ for bayesian inverse problems governed by pdes, IEEE

[Spantini et.al. 2015] Optimal low-rank approximations of Bayesian linear inverse problems, SIAM-SISC

[Polette 2025] Contributions to reduced basis methods for nonparametric Bayesian inference in geosciences, PhD

[.....]

Generalization to **nonlinear & nonGaussian** problems

$$\underset{v\in\mathbb{R}^d}{\arg\max}\frac{v^\top Hv}{v^\top\Sigma_{\text{pr}}^{-1}v} \qquad \text{where } H = ???$$

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- **[Cui et.al. 2016]**: Replace A with $\nabla u(X)$ and take posterior expectation

$$H(y) = \int \nabla u(x) \Sigma_{\text{obs}}^{-1} \nabla u(x)^\top \mathrm{d}\pi_{X|Y}(x|y)$$

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$$H(y) = \int \left(\nabla \log \mathcal{L}^y\right)\left(\nabla \log \mathcal{L}^y\right)^\top \, \mathrm{d}\pi_X$$

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- [Zahm et.al. 2022]: Use the **Fisher Information** matrix $\Rightarrow$ "certified" DR

$$\boxed{H(y) = \int \left(\nabla \log \mathcal{L}^y(x)\right) \left(\nabla \log \mathcal{L}^y(x)\right)^\top \mathrm{d}\pi_{X|Y}(x|y)}$$

Control the **Kullback-Leibler (KL)** [Zahm et al, 2022]

$$D_{\mathrm{KL}}(\pi_{X|y}||\widetilde{\pi}_{X|y}) = \int \log\left(\frac{\pi_{X|y}}{\widetilde{\pi}_{X|y}}\right) \mathrm{d}\pi_{X|y}, \quad \begin{cases} \pi_{X|y} \propto \mathcal{L}^y(x)\pi_X(x) \\ \widetilde{\pi}_{X|y} \propto \widetilde{\mathcal{L}}^y(U_m^\top x)\pi_X(x) \end{cases}$$

1

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For any U_m, we have

$$\boxed{\min_{\widetilde{\mathcal{L}}} D_{\mathrm{KL}}(\pi_{X|y}||\widetilde{\pi}_{X|y}) \leq \frac{\overline{\mathbb{C}}(\pi_X)}{2}\left(\mathbb{E}_{\pi_{X|y}}[\|\nabla \log \mathcal{L}^y\|^2] - \mathbb{E}_{\pi_{X|y}}[\|U_m^\top \nabla \log \mathcal{L}^y\|^2]\right)}$$

where $\overline{\mathbb{C}}(\pi_X)$ is now the **subspace Logarithmic Sobolev constant**[1].

[1] Log-Sob inequality: $\mathbb{E}[u(X)^2 \log(u(X))] \leq \mathbb{C}(\pi_X)\mathbb{E}[\|\nabla u(X)\|^2]$ for all $u \geq 0$ s.t. $\mathbb{E}[u(X)] = 1$

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Remark: similar to the Active Subspace bound

$$\boxed{\min_f \mathbb{E}[(u(X) - f(U_m^\top X))^2] = \overline{\mathbb{C}}(\pi_X)\left(\mathbb{E}[\|\nabla u(X)\|^2] - \mathbb{E}[\|U_m^\top \nabla u(X)\|^2]\right)}$$

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Minimizing the RHS corresponds to **perform a PCA on** $\nabla \log \mathcal{L}^y(X)|Y = y$

$\Rightarrow$ **Likelihood informed Subspace method** $\Leftarrow$

[1] Log-Sob inequality: $\mathbb{E}[u(X)^2 \log(u(X))] \leq \mathbb{C}(\pi_X)\mathbb{E}[\|\nabla u(X)\|^2]$ for all $u \geq 0$ s.t. $\mathbb{E}[u(X)] = 1$

How to **control** $\overline{\mathbb{C}}(\pi_X)$?

- **Bakry–Émery** theorem for log-concave π_X

$$\mathrm{Hess}(-\log \pi_X(x)) \succeq \rho I_d \qquad \Rightarrow \qquad \overline{\mathbb{C}}(\pi_X) \leq 1/\rho$$

- **Holley–Stroock** perturbation lemma

$$\alpha\, \rho_X(x) \leq \pi_X(x) \leq \beta\, \rho_X(x) \qquad \Rightarrow \qquad \overline{\mathbb{C}}(\pi_X) \leq \frac{\beta}{\alpha} C(\rho_X)$$

- Combination of the two above.

$$\boxed{\{\text{these assumptions}\} \Rightarrow \{(\text{subspace})\text{LogSob }\} \Rightarrow \{(\text{subspace})\text{Poincaré}\}}$$

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The **ideal setting** remains $\pi_X = \mathcal{N}(0, I_d)$. As for the (subspace)Poincaré inequality, it is preferable to **whiten** the prior

$$\overline{X} = \Sigma_{\mathrm{pr}}^{-1/2}(X - \mu_{\mathrm{pr}})$$

This is equivalent to computing the eigendecomposition of

$$\boxed{\overline{H} = \Sigma_{\mathrm{pr}}^{1/2}\, H\, \Sigma_{\mathrm{pr}}^{1/2}}$$

Other functional inequalities bound other divergences

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- (subspace)**Poincaré inequality** bounds **Hellinger distance** [Cui Tong 2022]

$$D_{\mathrm{Hell}}(\pi_{X|y}||\widetilde{\pi}_{X|y}) \leq 2(RHS)$$

Usefull *e.g.* for Laplace priors $\pi_X(x) = \prod_{i=1}^d \exp(-\lambda|x_i|)$ which do not satisfy the (subspace)LSI...

[Flock et.al. 2024] Certified coordinate selection for BIP with **Laplace prior**, Inverse Problems

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  This interpolates KL ($\alpha = 1$), Hellinger ($\alpha = 1/2$) and χ^2 ($\alpha = 2$)...

  [Li Marzouk Zahm 2024] Principal feature detection via Φ-**Sobolev inequalities**, Bernouli.

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- **Dimensional Logarithmic Sobolev inequality** improves the above bound

  $$D_{\mathrm{KL}}(\pi_{X|y}||\widetilde{\pi}_{X|y}) \leq \text{``}\log(1 + (RHS))\text{''}$$

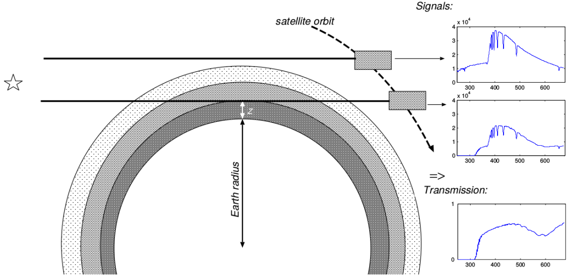
  [Li Cui Li Marzouk Zahm 2024] Sharp detection of LDS via **dimensional logSob inequalities**, IMA:I&I.

In practice...

Benchmark Gomos: atmospheric remote sensing

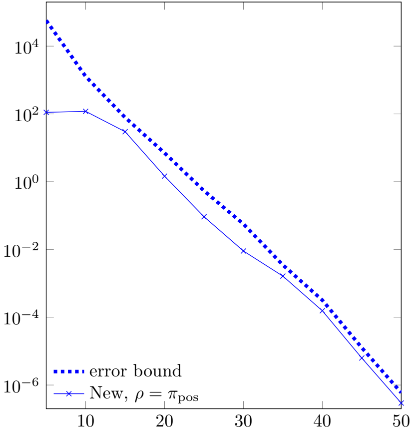
- [Haario Tamminen et al. 2004]
- Estimate gas densities $x = \rho^{\text{gas}}(z)$ from transmission spectra $y_\omega(z)$
- Beer's law:

$$y_\omega(z) = \exp\left(- \int_{\text{light path}} \sum_{\text{gas}} \alpha_\omega^{\text{gas}}(z(\zeta))\ \rho^{\text{gas}}(z(\zeta))\ \mathrm{d}\zeta \right) + \xi$$



- Gaussian prior $\mathcal{N}(\mu_{\text{pr}}, \Sigma_{\text{pr}})$ with squared exponential kernel covariance
- After discretization of the atmosphere, $\dim(x) = 200$.

$$D_{\mathrm{KL}}(\pi_{X|y}||\widetilde{\pi}_{X|y}) = \mathit{function}(r)$$



$$H(y) = \int (\nabla \log \mathcal{L}^y)(\nabla \log \mathcal{L}^y)^\top \, \mathrm{d}\pi_{X|y}$$

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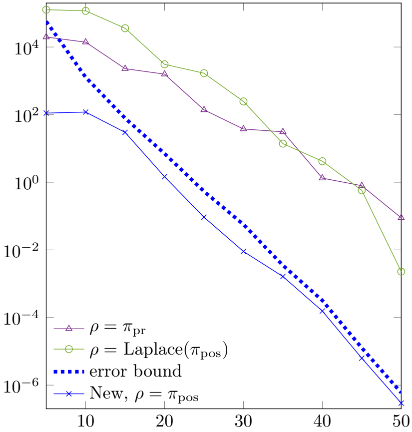

$$H^{(\rho)}(y) = \int (\nabla \log \mathcal{L}^y)(\nabla \log \mathcal{L}^y)^\top \, \mathrm{d}\rho$$

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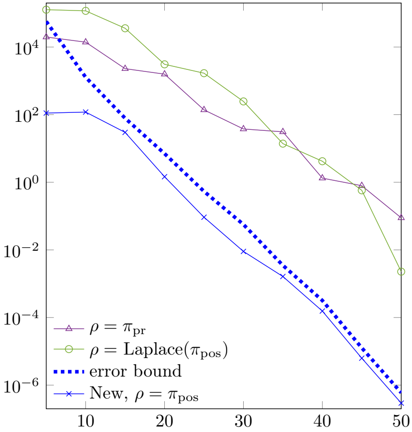

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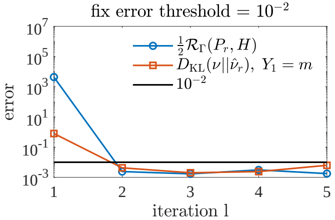
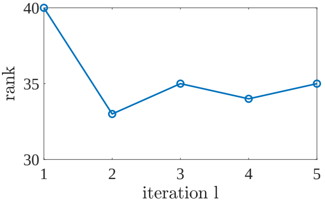
Iterative procedure:

$$\pi_X \to H^{(\pi_X)} \to \widetilde{\pi}_{X|y} \to H^{(\widetilde{\pi}_{X|y})} \to \dots$$





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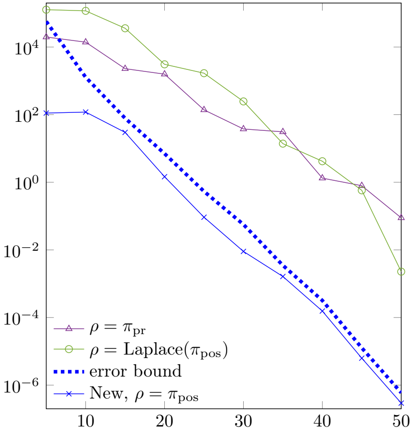

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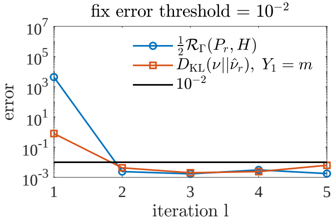

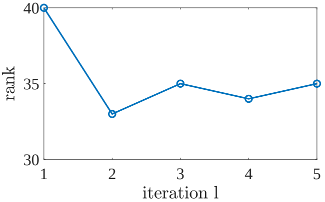

We have to re-run the whole procedure for every new piece of observed data y...

Data-free dimension reduction $U_m = \cancel{U_m(y)}$

Problem statement

Recall that, in the Bayesian perspective, the observed data y is a **realization** of a random variable

$$Y \sim \pi_{\text{data}}$$

Objective

Find a $U_m = \cancel{U_m(Y)}$ such that

$$D_{\text{KL}}(\pi_{X|Y} || \widetilde{\pi}_{X|Y}) \leq tol \tag{1}$$

in high probability (w.r.t. Y).

By Markov inequality,

$$\mathbb{E}_Y \left[D_{\text{KL}}(\pi_{X|Y} || \widetilde{\pi}_{X|Y}) \right] \leq \varepsilon$$

is sufficient to ensure (1) with probability greater than $1 - \varepsilon / tol$.

Online / Offline procedure

Offline

1. Compute
$$H = \mathbb{E}\big(H(Y)\big), \quad Y \sim \pi_Y$$
2. Solve the eigenvalue problem $Hu_i = \lambda_i u_i$ and let
$$U_m = [u_1, \dots, u_m] \in \mathbb{R}^{d \times m}$$

Online

3. Receive a realization y of Y,
4. Compute the optimal function $\widetilde{\mathcal{L}}^y(x_m) = \mathbb{E}_{\pi_X}\left(\mathcal{L}^y(X) \middle| U_m^\top X = x_m\right)$
5. Assemble the posterior approximation $\widetilde{\pi}_{X|y}(x) \propto \widetilde{\mathcal{L}}^y(U_m^T x)\pi_X(x)$

Proposition [Cui et.al. 2021]

The above procedure yields

$$\mathbb{E}\Big[D_{\mathrm{KL}}(\pi_{X|Y} || \widetilde{\pi}_{X|Y})\Big] \le \frac{\overline{\mathbb{C}}(\pi_X)}{2} \sum_{i=m+1}^{d} \lambda_i$$

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If **Gaussian prior**, "dimensional log-Sob" improvement [Li et.al. 2024]

$$\mathbb{E}\Big[D_{\mathrm{KL}}(\pi_{X|Y}||\widetilde{\pi}_{X|Y})\Big] \leq \frac{1}{2} \sum_{i=m+1}^{d} \log(1+\lambda_i)$$

How to compute $H = \mathbb{E}\left[H(Y)\right]$?

Proposition

$$H = \int \mathcal{I}(x)\,\mathrm{d}\pi_X(x)$$

where $\mathcal{I}(x)$ is the **Fisher information matrix** of the likelihood $\mathcal{L}^y(x) \propto \pi(y|x)$ defined by

$$\mathcal{I}(x) = \int \nabla \log \mathcal{L}^y(x) \nabla \log \mathcal{L}^y(x)^T \pi(y|x)\mathrm{d}y$$

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- **Gaussian likelihood:** $\mathcal{L}^y(x) = \exp(-\frac{1}{2}\|u(x) - y\|^2_{\Sigma_{\text{obs}}^{-1}})$

$$\boxed{H = \int \nabla u(x)^T \Sigma_{\text{obs}}^{-1} \nabla u(x) \, \mathrm{d}\pi_X(x)}$$

This corresponds to the **Active Subspace** on the forward model u!

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- **Poisson likelihood:** $\mathcal{L}^y(x) = \prod_{i=1}^m \frac{G_i(x)^{y_i}\exp(-G_i(x))}{y_i!}$

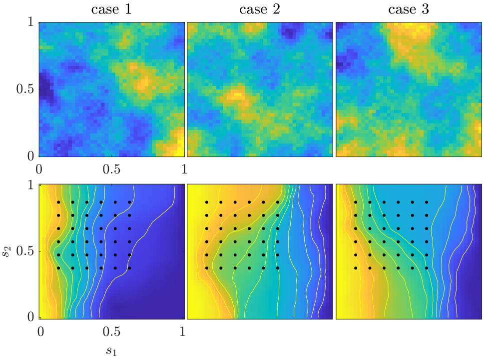
  $$\boxed{H = \int \nabla u(x)^T \mathrm{diag}\Big(u_1(x), \ldots, u_m(x)\Big)^{-1} \nabla u(x)^T\,\mathrm{d}\pi_X(x)}$$

- ...

A numerical example: elliptic PDE

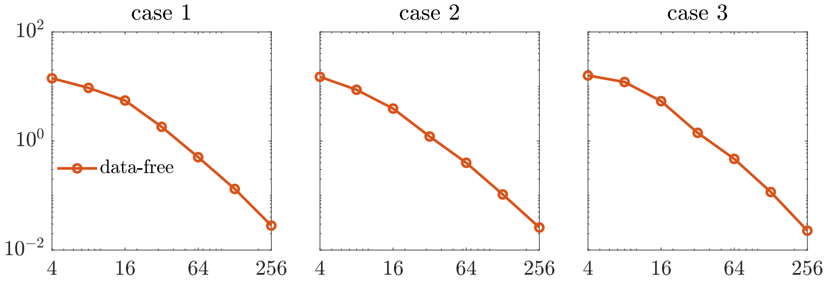
$$-\nabla \cdot \Big(\kappa(s)\nabla p(s)\Big) = f(s), \quad s \in [0,1]^2$$

- Parameter: $x = \log \kappa$
- Data: $y = (p(s_1), \dots, p(s_m)) + \mathcal{N}(0, \Gamma_{\text{obs}})$ (Gaussian likelihood)
- Gaussian prior: $-\Delta x + \gamma x = \mathcal{W}$ with $\mathcal{W}$ = white noise and $\gamma = 10$



Elliptic PDE: results for three different observations $Y^{(1)}, Y^{(2)}, Y^{(3)}$

$$D_{\mathrm{KL}}(\pi_{X|Y^{(i)}} || \widetilde{\pi}_{X|Y^{(i)}}) = \mathit{function}(m)$$



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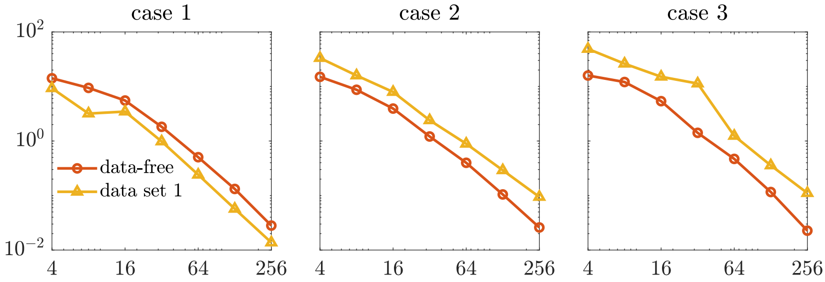

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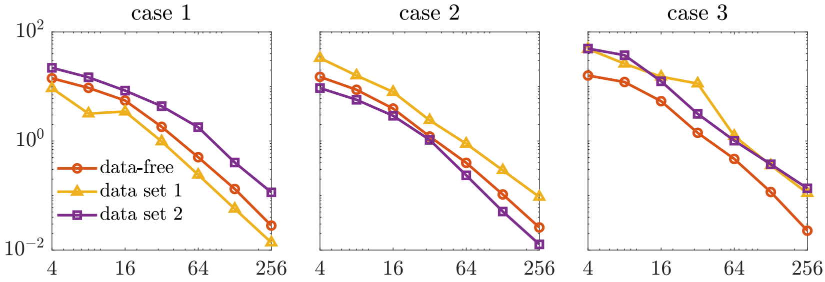

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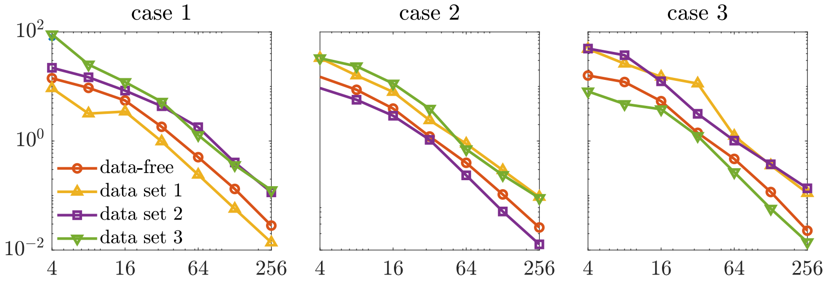

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- data set 3: U_m computed via $H(Y^{(3)})$

Conclusion

Certified likelihood-informed subspace

$$\widetilde{\pi}_{X|Y}(x|y) \propto \widetilde{\mathcal{L}}^y(U_m^\top x)\pi_X(x)$$

Gradient-based methods: (1) use functional inequalities to bound the error $\widetilde{\pi}_{X|Y} \approx \pi_{X|Y}$ and (2) minimize that bound over U_m using the diagnostic matrix

$$H(y) = \int (\nabla \log \mathcal{L}^y)(\nabla \log \mathcal{L}^y)^\top \, \mathrm{d}\pi_{X|y}$$

Many extensions:

- **Data-Free:** reduce X before observing the data
- **Parameter & Data reduction:** informed parameter & informative data
- **Rare event estimation:** nonsmooth likelihood: $\pi_X^\alpha(x) \propto 1_{u(x)\geq\alpha}\pi_X(x)$
- **Bounds on Conditional Mutual Information** to detect independence between X and Y given Z

[Zahm et.al. 2022] **Certified** dimension reduction in nonlinear Bayesian inverse problems, Math.Comp.
[Cui Zahm 2021] **Data-Free** Likelihood-Informed DR of Bayesian Inverse Problems, Inverse Problems.
[Cui Tong 2022] A unified performance analysis of likelihood informed subspace methods, Bernouli.
[Baptista et.al. 2024] Learning Non-Gaussian Graphical Models [...], JMLR
[Li Marzouk Zahm 2024] Principal feature detection via **Φ-Sobolev inequalities**, Bernouli.
[Li Cui Li Marzouk Zahm 2024] Sharp detection of LDS via **dimensional logSob inequalities**, IMA-II.
[Flock et.al. 2024] Certified coordinate selection for BIP with **Laplace prior**, Inverse Problems

Ongoing works on **optimal experimental design**

What data $Y_m = V_m^\top Y$ is informative on a parameter of interest $X_r = U_r^\top X$?

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- Bound the **Expectation Information Gain (EIG)**

$$\begin{aligned}\mathrm{EIG}(Y_m|X_r) &:= \mathbb{E}_Y[\mathrm{D}_{\mathrm{KL}}(\pi_{X_r|Y_m}||\pi_{X_r})] \\ &\geq \mathbb{E}_Y[\mathrm{D}_{\mathrm{KL}}(\pi_{X|Y}||\pi_X)] - \overline{C}(\pi_X)\mathbb{E}[\| V_m^\top \nabla u(X) U_r\|_F^2]\end{aligned}$$

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- **Refined bounds** on the EIG are comming (PhD of M. Doumbouya):

$$
\frac{1}{2}\ln\left(\frac{|V_m^\top \Sigma_{\mathrm{signal}}(U_r) V_m|}{|V_m^\top \Sigma_{\mathrm{noise}} V_m|}\right) \leq \mathrm{EIG}(Y_m|X_r) \leq \frac{1}{2}\ln\left(\frac{|V_m^\top \Sigma'_{\mathrm{signal}}(U_r) V_m|}{|V_m^\top \Sigma'_{\mathrm{noise}} V_m|}\right)
$$

Stay tuned!

Questions?